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Proximity and Localization

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TORRE A. and RALLET A. (2005) Proximity and localization, *Regional Studies* **39**, 47–59. The objective of this paper is to pave the way for an analysis of the relations between proximity and localization of activities and people, two notions that are often mistaken for one another. Our method consists in exploiting the semantic wealth of the notion of proximity. We distinguish two types of proximity (geographical and organized) and propose a grid of analysis of the main models of geographic organization of activities by articulating both types of proximity. We then introduce the phenomenon of tension between geographical and organized proximity in order to discuss problems that are often underestimated in spatial economy. First, organized proximity offers powerful mechanisms of long-distance coordination that constitute the foundation of the increasing geographical development of socio-economic interactions. The confusion between information interactions and knowledge exchange, and the constraint of being located in proximity neglects the fact that the collective rules and representations do manage, and at a distance, an increasing part of these interactions. It is then shown that there is a disjunction between the need for geographical proximity and co-localization of actors by introducing professional mobility and temporary geographical proximity. We also emphasize the ability of big organizations to manage the presence in different areas of their units, whereas smaller ones are more constrained by fixed co-localizations, which are only needed for certain phases of their interactions. Finally, we raise the often neglected question of the negative effects of geographical proximity, which creates tensions between the actors who use limited support-goods and tends to damage the local relational network. However, these negative effects can be limited by integrating them within organizations or institutions, that is through a re-composed organized proximity enabling one to solve conflicts and launch processes of cooperation or negotiation within ad-hoc mechanisms.

Geographical proximity Organised proximity Localization Ubiquity Interactions Organisation

TORRE A. et RALLET A. (2005) Proximité et localisation, *Regional Studies* **39**, 47–59. Cet article cherche à ouvrir la voie à une analyse des rapports entre la proximité et la localisation des activités et de la population, deux notions qui sont souvent confondues. L'approche consiste à exploiter la richesse sémantique de la notion de proximité. Dans un premier temps, on distingue deux types de proximité (géographique et organisée) et propose une grille qui permet une analyse des principaux modèles de l'organisation géographique des activités en articulant les deux types de proximité. Dans un deuxième temps on introduit le phénomène de la tension entre la proximité ou géographique, ou organisée, afin de discuter des problèmes qui sont souvent sous-estimés dans l'économie géographique. En premier, la proximité organisée fournit de puissants mécanismes de coordination à longue distance qui constituent la base du développement géographique croissant des interactions socio-économiques. La confusion entre l'interaction d'information et l'échange de connaissance, et la contrainte d'être situé à proximité ne fait aucune attention au fait que les réglementations et les représentations collectives gèrent, à distance, une proportion croissante de ces interactions. On démontre une disjonction entre le besoin de la proximité géographique et la localisation à proximité des agents, en introduisant la mobilité professionnelle et la proximité géographique temporaire. Il faut souligner aussi la capacité des grands établissements à gérer la présence de leurs unités sur des zones différentes, alors que de plus petits établissements se voient contraindre par des emplacements à proximité fixes dont ils n'ont besoin qu'à certaines étapes de leurs interactions. Pour finir, on remet en question les effets négatifs de la proximité géographique, souvent négligés, ce qui crée des tensions entre les agents qui emploient de façon limitée les services d'assistance technique, et a tendance à entraver le réseau relationnel local. Cependant, on peut minimiser ces effets négatifs-là tout en les insérant dans les organisations ou les institutions, par moyen d'une proximité organisée recomposée, ce qui permet la résolution des conflits et le lancement des processus de coopération ou de négociation à partir des mécanismes improvisés.

Proximité géographique Proximité organisée Ubiquité Interactions Organisation

TORRE A. und RALLET A. (2005) Nähe und örtliche Begrenzung, *Regional Studies* **39**, 47–59. Dieser Aufsatz setzt sich zum Ziel, den Weg zu einer Analyse der Beziehungen zwischen Nähe und örtlicher Begrenzung von Betätigungen und Menschen zu ebnet, als zwei Begriffen, die oft verwechselt werden. Die Methode stützt sich auf die Ausnutzung der vielfältigen semantischen Bedeutung des Begriffes 'Nähe'. 1. Es werden zwei Arten von Nähe unterschieden: geographische und organisatorische, und ein Analysennetz der Hauptmodelle geographischer Einteilung von Betätigungen durch klare Unterscheidung beider Arten Nähe vorgeschlagen. Dann wird das Phänomen der Spannung zwischen geographischer und organisatorischer Nähe eingeführt, um Probleme zu diskutieren, die in der Raumwirtschaft oft unterschätzt werden. 2. Zunächst bietet

organisierte Nähe wirksame Mechanismen der Fernkoordination, welche die Grundlage der zunehmend geographischen Entwicklung sozialwirtschaftlicher Wechselwirkung darstellen. Die Verwechslung von Informationswechselwirkungen mit Austausch von Fachwissen, und die Beschränkung, in benachbarten Standorten zu sein, vernachlässigt die Tatsache, daß gemeinsame Regeln und Darstellungen selbst auf Entfernung einen stetig zunehmenden Teil dieser Wechselwirkungen bewältigen. 3. Dann wird gezeigt, daß es eine Disjunktion zwischen dem Bedürfnis nach geographischer Nähe und gemeinsamer örtlicher Begrenzung der Spieler gibt, in dem professionelle Beweglichkeit und vorübergehende geographische Nähe eingeführt werden. Gleichzeitig wird die Fähigkeit großer Organisationen betont, deren Vorkommen auf verschiedenen Gebieten ihrer Abteilungen handzuhaben, während kleinere eher durch festgelegte, gemeinsame örtliche Begrenzungen eingeschränkt sind, die nur für bestimmte Phasen ihrer Wechselwirkungen benötigt werden. 4. Abschließend wird die oft vernachlässigte Frage nach den negativen Wirkungen geographischer Nähe aufgeworfen, die Spannungen zwischen den Spielern hervorruft, denen begrenzte Hilfsmittel zur Verfügung stehen, und dazu neigt, dem Netzwerk örtlicher Beziehungen zu schaden. Diese negativen Auswirkungen können jedoch durch deren Integration mit Organisationen oder Institutionen überwunden werden, d.h. durch eine neu zusammengesetzte, organisierte Nähe, die es gestattet, Konflikte zu lösen, und Wege der Zusammenarbeit oder Verhandlungen in ad-hoc Mechanismen zu beschreiten.

Geographische Nähe Organisierte Nähe Örtliche Begrenzung Verbreitung Wechselwirkungen
Organisierung

TORRE A. y RALLET A. (2005) Proximidad y localización, *Regional Studies* **39**, 47–59. El objetivo de este artículo es determinar el contexto para un análisis de las relaciones entre la proximidad y la localización de actividades y personas, dos nociones que a menudo se confunden. Nuestro método consiste en explotar la riqueza semántica que existe en torno a la noción de proximidad. Distinguimos dos tipos distintos de proximidad (geográfica y organizacional), y proponemos un análisis de los principales modelos de organización geográfica de actividades mediante la articulación de ambos tipos de proximidad. A continuación introducimos el fenómeno de tensión en la proximidad geográfica y la organizacional con el fin de examinar una serie de problemas que a menudo se ven sub-estimados en la economía espacial. En primer lugar, la proximidad organizacional ofrece poderosos mecanismos de coordinación a larga distancia que constituyen el fundamento del creciente desarrollo geográfico de las interacciones socio-económicas. La confusión entre interacciones de información e intercambio de conocimiento y el obstáculo que supone estar ubicado en proximidad deja de lado el hecho de que las reglas y las representaciones colectivas manipulan, y a distancia, una parte cada vez mayor de dichas interacciones. A continuación mostramos que existe una disyunción entre la necesidad de proximidad geográfica y la colocalización de actores, mediante la introducción de movilidad profesional y proximidad geográfica temporal. También destacamos la habilidad que tienen las grandes organizaciones para manejar la presencia de sus unidades en diferentes zonas, mientras que las más pequeñas se ven restringidas en mayor medida por colocalizaciones que se necesitan solamente para ciertas fases de sus interacciones. Por último, planteamos una de las preguntas que a menudo se ha dejado de lado sobre los efectos negativos de la proximidad geográfica, lo cual crea tensiones entre los actores que hacen uso de un número limitado de bienes de apoyo, y tiende a perjudicar el *network* relacional local. No obstante, estos efectos negativos se pueden aminorar si se integran en las organizaciones o instituciones, es decir, mediante una proximidad organizacional recompuesta permitiéndole a uno solucionar conflictos y lanzar procesos de cooperación o negociación dentro de mecanismos *ad hoc*.

Proximidad geográfica Proximidad organizacional Localización Ubicuidad Organización

JEL classifications: L0, O3, Q0, R1

INTRODUCTION

In an ever more globalized economy marked by the increasing nomadism of firms (ZIMMERMAN *et al.*, 1995) and the mobility of individuals, proximity still matters. Territories are being rediscovered, local systems given more importance, the merits of the decentralization of decision-making are celebrated and most people agree that more decisions should be made at the local level. The origin of this phenomenon is clear: the more globalized are relations, the more anonymous they become and the more uncertainty they create; this leads to the necessity for economic actors to lean against local identities, support bases thanks to which they can face international competition. It is therefore hardly surprising to see the term 'proximity' so widely used in the contemporary discourse, in particular that of developers, planners and politicians.

But what is proximity? Or, more precisely, what is its relation to geographical space? The question might seem preposterous: does not 'being in proximity' simply mean being situated near and not somewhere else? Yet, this answer is not that obvious. What does 'being near' somebody mean? What does it imply in social and economic relations? Neighbours might ignore or even hate one another. Local firms can be rivals and refuse any cooperation. Furthermore, one can be present and active both here and there thanks to communication technologies and travel. Finally, being in proximity of someone not only means being near him/her, but also means having a strong complicity with a person who is geographically distant, whether that person belongs to the same circle of friends, family, or even to the same network of firms or professionals.

Therefore, the term 'proximity' is much more ambiguous than the term 'localization'. But at the same

time, it is this ambiguity that makes the term interesting because it concentrates in a single term the multiplicity of spatial scales within which economic actors and individuals situate their actions. The term should therefore be retained, but its ambiguity should be lifted by exploring and developing the different forms of proximity. In particular, the multiplicity of proximity should help one shed new light on the questions of localization and geographic concentration and thus go beyond the traditional literature that generally only considers mono-located actors and monopolar spaces.¹

The objective of the paper is to propose an analysis of the relations between proximity and localization based on a basic grid of analysis progressively developed. The first section presents the grid of analysis based on the duality of proximity. Two types of proximity (geographical and organized) are defined for this purpose. The articulation of both types of proximity provides a simple but relevant grid of analysis of the way in which economic actors are nowadays 'situated' in geographical space. Using the grid of analysis, the second section discusses the confusion, which is frequent in the literature, between agglomeration and interactions of geographical proximity. The importance of the phenomena of social and institutional embeddedness (GRANOVETTER, 1973) is stressed to account for the role of geographical proximity in economic coordination, and it is shown that one of the most important questions of research is to explain the phenomenon of ubiquity that characterizes nowadays the behaviour of economic actors. The third section introduces the mobility of individuals. Indeed, one often forgets that the need for geographical proximity (to realize cooperation, for instance) does not necessarily imply that the actors should be located near each other. Indeed, it is increasingly less the case. For people travelling frequently or staying away for temporary periods, the proximity constraint becomes relative. Finally, geographical proximity does not only have virtues, even though the literature generally only stresses these dimensions. It can be a factor of conflict, rivalry or negative externalities, which the mobilization of resources of organized proximity can help solve (the fourth section).

GEOGRAPHICAL PROXIMITY AND ORGANIZED PROXIMITY: A FRAMEWORK OF ANALYSIS

Various definitions of proximity, referring to notions such as space, neighbourhood, institutions, inter-individual relationships, epistemic communities, kinship and representations, have been offered in the literature of economics, geography, mathematics and sociology (BELLET *et al.*, 1998). In the field of economics, BELLET *et al.* (1993) and KIRAT and LUNG (1999) refer to three types of proximity (institutional, organizational, geographical), whereas other authors, such as

BOSCHMA (2003) or HURIOT (1998), identify more categories that, according to them, correspond to various types of relationships between individuals or institutions. The present authors have retained a simple definition based on a distinction between two types of proximity, called geographical proximity and organized proximity, respectively (TORRE and GILLY, 1999; RALLET and TORRE, 2000; RALLET, 2002).

As its name indicates, the former is a geographic notion, whereas the latter is not of a geographic nature. It is the nature of both types of proximity that is at the basis of their distinction. They are now examined.

- Geographical proximity expresses the kilometric distance that separates two units (e.g. individuals, organizations, towns) in geographical space. It has two essential properties. It is binary: naturally, there exist infinite gradations (more or less far from, more or less close to), but the purpose of examining geographical proximity is to determine whether one is 'far from' or 'close to'. A series of consequences, presented below, ensues from this division. It is doubly relative. First, geographical proximity, which is at the basis of the division between proximity and distance, is relative to the means of transport. The kilometric distance is weighted by the time and/or cost of transport. Second, proximity is not only an objective data. It proceeds from a *judgement* made by individuals on the nature of the geographical distance that separates them. The judgement consists in processing the parameters that influence the distance, to convert them into the statement according to which one is close to or far from. These parameters include objective data (km, time, price), but also the perception individuals have of them. And this perception varies, for example, according to the age, social background, gender, profession of people. (For example, the possibility of meeting someone once a day can be perceived differently according to individual.) However, although it has a social (determined by the means of transport) and subjective (referring to a statement) nature, geographical proximity may be, at a time *t*, considered as a physical space data representing a constraint imposed, at that particular time, on the actors to develop their actions. This is how the present authors understand the term.
- Organized proximity is not geographic but relational. By organized proximity, the ability of an organization² to make its members interact is meant. The organization facilitates interactions within it and makes them a priori easier than with units situated outside the organization. Two main reasons explain this. First, belonging to an organization translates into the existence of interactions between its members, inscribed – as the evolutionist language puts it – in the *genes* (routines) of the organization. This is what is called here the *logic of belonging* of organized proximity: two members of one organization are

close to each other because they interact, and because their interactions are facilitated by (explicit or implicit) rules and routines of behaviour that they follow. Thus, other things being equal, cooperation will, *a priori*, develop more easily between researchers and engineers belonging to the same firm, the same technological consortium or the same innovations network. Second, the members of an organization are said to share a same system of representations, or set of beliefs,³ and the same knowledge. This social relation is mainly tacit. This is what is called here the *logic of similarity* of organized proximity. Two individuals are considered as close because they are 'alike', i.e. they share a same system of representations, which facilitates their ability to interact. Thus, two researchers belonging to the same scientific community will cooperate more easily because they share not only the same language, but also the same system of interpretation of texts, results, etc. These two logics are partly complementary, partly substitutable. They are complementary because the shared beliefs (or *cognitive maps*) limit the possible divergent interpretations of the formal rules and thus makes coordination effective through rules. Similarly, interactions founded on tacit representations generally lean on a minimum number of formal rules (e.g. conventions or contracts between scientific laboratories). But the *logic of belonging* and the *logic of similarity* are also partly substitutable: in an informal community, i.e. an organization with no strong explicit rules, the weakness of the cooperation between the members can be compensated for by the existence of strong behavioural cohesion that creates implicit rules of interaction. It is the case of communities of researchers that are not highly structured formally but which are characterized by a strong cohesion resulting from the homogeneous formation of a university group of researchers at international level.

The intersection of both types of proximity (geographical and organized) provides a grid of analysis of the different models of geographic organizations of activities. Thus, industrial districts, innovation milieus or localized systems of production (LSP) are characterized by the existence of both types of proximity. In this type of model, organized proximity – defined by the intensity of the client–supplier relationships, the exchange of know how or the existence of an 'industrial atmosphere' – is based on the co-localization of actors within a determined zone.

Although this model is widely discussed in the economic literature, it is but one model among others. Indeed, organized proximity – which consists of functional relations (interactions) or relations between people sharing the same 'identity' (common beliefs and cognitive maps) founded on the organization and not on the territory – often exists without any geographical

proximity. In this case, geographical proximity will be weakly organized, if at all, whereas organized proximity will not take any geographic dimensions. Because organization is not geographic in essence, it has the ability to cross territories and frontiers. It is situated in a space, takes territories into account, but is not defined or limited by them: the organization of a multinational firm is a good example.

This point is important because one often neglects to distinguish the *territorial genealogy* of organizations (any organization has a territorial origin: it is founded and develops originally on one given territory) from *their nature which is a-territorial* (the interactions, the explicit or implicit rules that define it). The innovative milieus (MAILLAT, 1995) and industrial districts (ASHEIM, 1996) approach has introduced great confusion in this regard by using a particular case as an analytical model. The approach adopted here does the opposite: it is not based on the fusion of both types of proximity, but on their disjunction. The first approach (the fusion of both types of proximity as an analytical model) confines the analysis of local development to a specific case *de facto* considered as a norm. The authors of the districts or milieus movement admit that this norm is not universal and that there are other possible models of territorial development, but their analytical mechanism, trapped in the fusion of both types of proximity, cannot account for it. The second approach (the disjunction of both types of proximity) straightaway introduces the plurality of the forms of territorial development and proposes a method to examine them: analysing how two types of proximity that are analytically different (geographical and organized) are articulated.

To illustrate this idea, Table 1 has been created of the relations between geographical proximity and organized proximity. Read from left to right, it highlights the results obtained when both types of proximity

Table 1. *Intersection of both types of proximity and its results in terms of interactions*

	Geographical proximity	Organized proximity
Geographical proximity	Agglomeration without interactions: I have a lot of neighbours but I do not know them (no direct coordination) Agglomeration due to infrastructures for example	Local systems of innovation, production, clusters . . . : Geographical proximity activated by organized proximity
Organized proximity	Non-permanent co-localization: Long-distance coordination implies temporary face-to-face (sales representatives, researchers)	Supra-local organizations: Coordination between sedentary people located in different places, supported by organizational rules and information and communication technologies

cross each other and the consequences of this intersection in terms of interactions between firms at local level. The top left box means that geographical proximity alone cannot generate synergies, and that it cannot create interactions between economic actors at the local level. Geographical proximity facilitates interactions (e.g. random meetings) and enables actors to observe the actions undertaken by others and to draw comparisons (MALMBERG and MASKELL, 2002), but it does not in itself facilitate coordination. When geographical proximity only crosses itself, economic actors are agglomerated but have no direct relations with one another. This case enables one to distinguish agglomeration from localized relations, which the empirical literature too often omits to do (for some works on geographic spillovers, see JAFFE *et al.*, 1993; WALLSTEN, 2001).

Indeed, one cannot infer from the co-localization of actors that they necessarily have direct relations with each other and that it is for this reason that they are in the same location. On the contrary, it is important to show that the current phenomena of agglomeration, in particular in big cities, is not necessarily based on direct functional relations because of the difficulty of establishing these relations in these big centres. Great cities function like hubs paradoxically making long-distance access easier than local access. In this case, the economic actors agglomerate geographically, not because of a direct need for coordination, but because they have similar access to infrastructures (e.g. airport, high-speed railway station, highways, interchanges).

The top right box shows that to generate interactions, geographical proximity must be structured and activated by organized proximity. It is the case of the above-mentioned districts, milieus and other local systems of production or innovation. Similarly, the negative effects of geographical proximity can be overcome by mobilizing the resources provided by organized proximity. This case will be examined in the fourth section.

The bottom left box indicates that organized proximity can be transformed temporarily into geographical proximity, or even better, that it can only survive at a long distance through the implementation of temporary meetings using geographical proximity (the situation of sales representatives or business trips in the context of a cooperation project between firms). As in the previous case, organized and geographical proximities are complementary but only temporarily so. This case will be examined in the third section.

Finally, the bottom right box illustrates situations in which supra-local organized relations occur: e.g. multi-unit firms, global networks of firms, national or international professional communities. The supports of coordination are the sharing of norms and standards (such as ISO 9000 standards), the existence of formal rules and common representations. There is little or no individual mobility because coordination does not

require face-to-face interactions. It is the case when, before the realization of a collective project, tasks have been precisely divided between geographically dispersed work units, and when coordination only concerns technical aspects and no decisions have to be made. The creation of a software product whose different modules are designed by several teams located on different continents is a good example of this.

The intersection of the two analytically distinct types of proximity has enabled several models of geographic organization of activities to be distinguished. It also enables one to relativize the implicit postulate underlying most spatio-economic analyses, i.e. the search for geographical proximity as a factor of localization of firms and households.

FROM THE IMPORTANCE OF SOCIAL NETWORKS AND INSTITUTIONS TO THE UBIQUITY OF ACTORS

It is very tempting to explain the process of spatial concentration of firms and populations – which is one of the major characteristics of contemporary economies – by the existence of direct externalities of proximity.⁴ This explanation is not exclusive of the industrial districts and innovative milieus movement. It is found in economic literature as a whole.

According to this explanation, actors concentrate in the same locations because geographical proximity is necessary for them to interact. This is one of the answers given by LUCAS (1988) to the question of the localization of industrial activities, or by the New Economic Geography, which considers that economic actors (firms and workers/consumers) tend to agglomerate because they have relations of exchange of goods or work (KRUGMAN, 1991). Other authors highlight the transfer of knowledge or information to explain co-localization (see literature on the geography of innovation: FELDMAN, 1999; FELDMAN and MASSARD, 2002), or simply psychological or sociological reasons leading them to settle closer to one another, pushed by a need of a social nature (OTA and FUJITA, 1993). Whatever the motive (financial, technological, social), externalities are said to be externalities of proximity. But this is seldom demonstrated. More often than not, it is merely an implicit claim: indeed, as MALMBERG and MASKELL (2002) have shown, most empirical works attempt to prove the existence of geographic spillovers rather than to produce empirical evidence at firm level. In these approaches, the search for geographical proximity only appears economically limited through congestion forces that prove to be the negative aspect of agglomeration (increase of land, property, labour, environmental, social costs).

This answer, which mistakes the search for geographical proximity and the process of localization, is invalidated if one observes the evolutions of the contemporary

world. Without mentioning globalization, it is quite clear that the geographic sphere of interactions between economic and social actors has considerably widened in the last few years, whether they be of individuals or organizations. The search for permanent geographical proximity is no longer the factor generally emphasized in the strategies of firms, in particular of big firms, the units of which are sometimes called *footloose* (free from any geographical attachment). Similarly, the need for individuals to be mobile has significantly increased (leisure travel, mobility to go to work or job changes and the modifications of localization they imply). This process is even true for actors said to have a limited spatial horizon such as small- and medium-sized enterprises (SMEs) (which are developing their ability to find suppliers or markets far away) or employees (through the increase of interregional geographic mobility or the important increase in the distance between home and the work place).

And yet, in spite of these obvious facts, the existence of direct externalities of geographical proximity seems to be considered as a *fact of nature* in the literature dedicated to spatial questions (the literature that considers that face-to-face interactions will always be required to establish social contracts and exchange information). This approach has been adopted not only by the new economic geography, but also by more heterodox works that justify the co-localization of innovating firms by the natural properties of knowledge (with equations of the type: tacit knowledge = face-to-face transmission = need for geographical proximity = constraint of co-localization) (HOWELLS, 2002). This presupposition, often given as a basic postulate of the analysis, remains to be proved. At the very least, one should question the role of interactions of proximity in an economy founded on the increasing de-territorialization of economic relations or – as it is called in the present grid of analysis – the increasing disjunction of geographical proximity and organized proximity (development of the bottom right box of Table 1).

Thus, it appears judicious to relativize these analyses by integrating behaviours other than the mere search for geographical proximity leading to a process of localization. First, the search for geographical proximity does not always lead to a process of localization and, second, the development of long-distance interactions thanks to the increasing mobility of individuals and information tends to reduce the weight of local coordination. In particular, the increase of non-local organized proximity tends to demonstrate that local development is not exclusively founded on the search for synergies between local actors.

In this regard, two main ideas have emerged from the research carried out by the group 'Proximity Dynamics':⁵

(1) Although for certain types of activities of production or transactions, the constraint of geographical proximity subsists, it is highly relative in economic

coordination, including for activities supposed to require it, such as information and knowledge-intensive activities (RALLET and TORRE, 2000).

In any case, the need for geographical proximity in coordination cannot alone explain the geographic concentration of actors and the existence of systems of production or innovation with a local (or partly local) basis.

Two other factors explain the relative nature of the constraint of geographical proximity. First, economic relations (for which the need for geographical proximity is reduced) are embedded in highly territorialized social networks (GERTLER, 2003). In this perspective, the existence of localized networks of innovation is less due to the functional need for face-to-face relations in order to exchange knowledge than to the fact that cooperation occurs between researchers and engineers belonging to different organizations but originating from the same university or belonging to the same social and family network (GROSSETTI and BÈS, 2001). Geographical proximity is not so much an *economic cause* of agglomeration as a *social effect* of the embeddedness of economic relations in inter-individual relations. Face-to-face interaction between two actors cannot alone generate synergies; the latter can only develop between two individuals who belong to the same network or share common representations.

Second, the geographic framework of economic interactions is largely conditioned by the role of institutions. Nowadays, geographical proximity appears to be a powerful factor of legitimacy of these institutions (valorization of the local in itself). Thus, local policies produce geographical proximity institutionally as a privileged mode of economic interactions. The search for synergies between local actors has logically become the alpha and omega of most policies of local development.

Therefore, if geographical proximity is given so much value in the discourse of local development, it is less for reasons that are intrinsic to the need for economic coordination than because of a double embeddedness of economic interactions in social networks on the one hand, and in institutions on the other hand. These conclusions are in keeping with the research orientations shared by a certain number of current movements in the economic analysis (interactionist, neo-institutionalist, evolutionist, regulationist) inasmuch as – as KIRMAN (1999) puts it – interactions between economic actors strongly depend on the organizational structure of the economy. The analysis of the articulation between this framework and economic interactions then becomes indispensable to understand the spatial dimension of coordination.

(2) What is at stake is not so much to determine whether long-distance coordination is going to replace interactions of geographical proximity or whether local relations will prevail. It is rather to show *the diversity of spatial scales* to which actors establish their interactions.

Nowadays social and economic actors are often in a situation of ubiquity, i.e. capable of being at once here and there

Far from being a product of science fiction, this capacity, which has existed since the invention of the telephone, has increased decisively since the emergence of the Internet and the development of information and communication technologies (ICT). One individual, or even better, a firm, can act at once locally and globally by making its suppliers compete with each other at global level, or by passing orders on stock exchanges abroad, for example. Actors are not only localized, but also capable of acting in real time in different places, which means that their registers of actions go far beyond their mere location and that they can develop interactions at local and 'global' scales (which has been possible for a long time with the development of techniques of transport) at the same time, in real time (which is new). This characteristic of actors' behaviours concerns not only the debate on firms' ability to acquire competencies outside their local system of production, but also the relations between all individuals capable of acting at once here and there. It is this relative ubiquity of the action of actors in geographic space that must be examined to understand how a 'space of flows' is linked to a 'space of places' (as mentioned in archipelago economics; VELTZ, 1996).

This approach of proximity, a combination of geographical proximity and organized proximity taking into account the ubiquity of actors, questions once again the traditional conception of the localization of activities. The problem is no longer to determine where an actor is localized (in relation to the localization of other actors), but to understand how the action of actors develops simultaneously at different spatial scales. Furthermore, the localization of actors must be distinguished from geographical proximity, which can be obtained thanks to more temporary meetings.

TEMPORARY NEEDS FOR GEOGRAPHICAL PROXIMITY

Another way of showing the necessity of separating search for geographical proximity from co-localization of individuals and organizations is to emphasize the importance of the increasing mobility of men, information and goods. Take the example of man's mobility. The importance of this mobility is underestimated in the literature. For example, it is often claimed that the localization of a service-providing firm in a given area means that its employees actually work in this area. But these services are increasingly more mobile: a service is often provided to clients situated in an area other than that to which the employee is attached administratively. The increasing nomadism of employees tends to separate the place where they actually work from the place to which they are attached administratively. In France, in the last 20 years, the relative part of the

distances between people's home and their fixed place of work has decreased because of the generalization of the continuous workday, but also because the part of mobility between the various places of work has increased (ORFEUIL *et al.*, 2000). This is due to the increasing part of mobile workers.

Indeed, the professional mobility of individuals has increased with the development of transports (improved accessibility, increase of speed, reduction of costs) and the technological revolution in telecommunications (improved forms of long-distance processing and transfer of information in comparison with the telephone era, low costs of information transfer). The complementarity of transport and communication (the more individuals telecommute, the more they need to meet others, and vice versa) increases this mobility, so that an increasing number of actors no longer have a permanent work place. They work by travelling, i.e. not where they officially work according to statistics. Thus, in France, the number of people working outside their home in a fixed location dropped by 4% between 1982 and 1994, and the number of people working in different locations increased by 600 000 during the same period. Also the number of workers travelling as part of their work to various locations within the region where they are employed (i.e. short-distance trips) increased by 40% between 1982 and 1994.

Apart from migrations, which are not included in the present field of analysis, spatial economy essentially considers one type of mobility: the daily mobility of employees between their home and work (commuting), but this type of mobility hardly questions the traditional conception of territories. It is actually used to delimit one type of territory (zones of employment defined by alternating migrations). But there are wider mobilities that cross territories and contribute to dis-identify them: the travelling of a sales representative, the round of duty of a maintenance engineer, the visits of several days of a consultant auditing a firm, the weekly or annual touristic trips of a family, the participation of a researcher to a national or international colloquium, the temporary visit of an engineer to the laboratory of a firm or university with which his firm cooperates (for the long-distance work-related mobility among employees in Danish high-competence organizations).

Thanks to these developing mobilities, the constraint of geographical proximity, which is real for certain types of interactions – in particular for services or the sharing of knowledge – can be fulfilled temporarily through travelling without the interaction leading to the permanent co-localization of the partners.

In particular, one must emphasize the fact that the need for geographical proximity is generally not permanent. It affects certain phases of the interaction: the phase of negotiation in a transaction, the definition of guidelines and the organizational framework of cooperation, the realization of its initial phase in the case of a technological alliance, the necessity to share

equipment in the experimental phase of a common research project or to exchange knowledge and, above all, to know personally the researchers (colloquium) belonging to a scientific community. Short- or medium-term visits are then sufficient for the partners to exchange – during face-to-face meetings – the information needed for cooperation. As a result, permanent co-localization is not necessary even for activities where physical interaction plays an important role in the coordination (services co-produced by the provider and the user, knowledge-intensive activities such as innovation and research and development (R&D) activities). This is what are called *the need for temporary geographical proximity*.

The temporary nature of the need for proximity has an impact on the localization of firms. Indeed, the possibility of moments of temporary proximity tends to relativize one of the most widespread theses in the regional analysis, i.e. the fact that firms have a strong tendency to settle near one another because of frequent and repetitive interactions requiring face-to-face relations. This idea can be found, in particular, in the research carried out in the field of innovation geography (FELDMAN, 1999). According to some authors, firms need geographical proximity to exchange knowledge concerning their production, commercialization and, above all, R&D activities. The thesis is based on the tacit nature of part of the knowledge, the transmission of which relies on face-to-face relations (e.g. learning by imitation, informal exchanges, intuitive solutions to problems), whereas codified knowledge is transmitted more easily through ICTs or physical supports (e.g. papers, books, instruction manuals), which are independent from the individuals or organizations that produced them.

In the context of the present research (KIRAT and LUNG, 1999; RALLET and TORRE, 2000; GROSSETTI and BÈS, 2001), this thesis has been discussed and relativized. The equation of the sharing of tacit knowledge and geographical proximity on the one hand, and codified knowledge and long-distance relations on the other hand, is indeed simplistic. First, it is difficult to separate the uses of both types of knowledge and therefore to translate them with different geographical terms. Second, face-to-face relations, and therefore geographical proximity, are not the only possible support for the sharing of tacit knowledge (FREEL, 2003; BATHELT *et al.*, 2004). Thanks to the collective rules and representations they produce, organizations offer powerful mechanisms of long-distance coordination (organized proximity). The movement of globalization depends precisely on the capacity of organizations to widen the geographic sphere of interactions. Third, ICTs also make the long-distance sharing or co-producing of tacit knowledge possible thanks to the technological evolution of computer sciences that offers possibilities such as informal or visual communication (association of the image, written support and voice)

or written communication that has become close to oral communication (e.g. e-mails, forums, chats).

There is no denying that face-to-face relations remain indispensable for certain types of interactions, in particular when solving problems related to the heterogeneity of reasoning modes or those related to the processes of deliberation and negotiation. One could mention the example of two actors who do not know each other and who start cooperating on a new programme or new technology. However, the intensity of the need for face-to-face relations varies according to the phase of the process, as shown by the example of transfers of technology in the sector of biotechnologies (GALLAUD and TORRE, 2005). In this sector, the cooperations between firms consist of successive phases that condition their relation to space. The role played by geographical proximity diminishes with time. It is used in complementarity with organized proximity during the phase of co-production of fundamental, tacit and contextual knowledge. Its role diminishes subsequently during the phase of absorption of the knowledge produced during the scientific phase, which implies a re-contextualization of the latter in order to test it in various situations. Finally, it is often replaced entirely by organized proximity in the phases dedicated to the design of prototypes and clinical trials or to the codification of research results. Only two types of situations necessitate face-to-face interactions:

- Launch of innovative projects, in particular in cases where the actors have very different knowledge bases and where the project is not very structured (RALLET and TORRE, 2000).
- Cases of conflict management between innovators, proximity facilitating consultation between the participants as regards the use of communication facilities.

The needs for permanent or temporary geographical proximity are then different according to whether the firm faces a problem related to a choice of location or to the search for a new partner for an innovative project (GALLAUD and TORRE, 2005):

- In the case of firms entering a new sector, the choice of location varies according to the size of the enterprises. Large, multi-unit enterprises can choose to set up their R&D department in a single, large laboratory, thus benefiting from scale economies and avoiding the duplication of research programmes; or they might prefer to create several small laboratories in proximity of their main clients in order to understand more quickly the latter's needs. SMEs, however, because of limited financial and human resources, have to locate in proximity of other firms or organizations whose role in terms of innovation is crucial.

- In the case of firms that are already established and which are not necessarily located in proximity of organizations with which they wish to cooperate on innovation, there are several possibilities: create a joint venture or relocate the staff in charge of innovation projects, either for the whole duration of the projects or only for short periods. The first solution is relatively infrequently chosen by firms because it is very expensive. The second solution has the advantage of being flexible; but relocating staff is equivalent to depriving the firm of part of its resources for the duration of the project, which proves difficult for small enterprises. The most frequently chosen solution consists in simultaneously assigning some employees to different projects located in different areas, the employees in question travelling to the different locations when necessary. Surveys on inter-firm cooperations show that in most cases, firms cooperate with organizations not located in the same regions (FREEL, 2003; TETHER, 2002).

Indeed, the bigger the firm, the more easily it adjusts its localizations to the temporal nature (e.g. permanent, temporary) of the need for geographical proximity. Thus, big firms can fulfil more easily the need for geographical proximity by de-localizing part of their staff, including for relatively long periods, whereas smaller firms (very small enterprises or small SMEs) are often forced to adopt a permanent co-localization, even when they only need temporary geographical proximity. Big firms, group subsidiaries or universities can bypass the constraint of co-localization associated with the initial phase of exploration by sending teams of researchers or doctors for short or prolonged visits to distant research centres, for example. These solutions are possible thanks to the important volume of human resources available to them. However, in the case of smaller organizations, the coincidence between the need for R&D exploration and the need for a permanent geographical proximity during this process is often a determining factor of localization, one person being appointed to tasks that are part of different phases of the R&D process. They are then forced to settle near other firms or laboratories, even if they only need geographical proximity during one phase of their R&D process.

Therefore, geographical proximity must not be mistaken for localization of firms, the existence of permanent geographical proximity during the phase of R&D exploration in no way implying a compulsory localization in proximity, whereas these data often represent a constraint of localization for smaller firms. This is one of the reasons why the networks of innovations are highly localized in the case of small firms, which is less true for bigger firms.

The analysis of geographic mobility, of its increase, of its various forms and uses by firms depending on their size and by individuals depending on their income

represents a great challenge for spatial economy. The present authors believe that the new patterns of spatial organization of activities cannot be explained without such an analysis. In any case, the latter reveals that mistaking a need for geographical proximity with localization is hazardous, if only because part of these needs can be met thanks to temporary geographical proximity.

FROM THE NEGATIVE DIMENSIONS OF GEOGRAPHICAL PROXIMITY TO THE RESTORING QUALITIES OF ORGANIZED PROXIMITY

Mobility is a factor that is essential to the functioning of contemporary economies, but this does not mean that all the components of economic systems are necessarily mobile: actors, goods, infrastructures or natural resources can prove totally or partly immobile for reasons related to natural (land availability), social (attachment to a territory of origin) or economic (financial limitations) constraints. Localization is then highly constrained by geographical proximity. It is the case of many activities related to the exploitation of the ground or underground, and therefore of an important part of agricultural and agri-food activities for which the relation to the land remains essential and which are therefore confined to a given territory (situation of *spatial lock in*), at the risk of having to share it with undesirable neighbours. Such tensions emerge when occupying the land forces actors to locate close to other actors who do not share the same logics of belonging or similarity.

More generally, tensions and conflicts emerge when using and occupying a piece of land imposes a localization close to other actors (the actors then suffer a constraint of geographical proximity) who do not have the same logics of belonging or similarity (the drawbacks of the constraint of geographical proximity can partly be overcome by mobilizing the virtues of organized proximity). There again, one finds, in a slightly different role, the geographical proximity – organized proximity diptych:

- Geographical proximity is a source of tensions at local level (also highlighted by BOSCHMA, 2003).
- Organized proximity can be mobilized to overcome these tensions and conflicts (in the sense of SCHELLING, 1960) through processes of cooperation or negotiation.

The situation of *geographical proximity constraint* is related to the presence of *support goods*, generally land or water. It creates negative externalities of proximity partly due to the localization of actors on one same support-good. These negative externalities may include the damage caused by eroding runoff, diffuse pollution or toxic emissions provoked by the presence of industrial activities in the proximity of population concentrations. But this localization can also provoke conflicts related to

the question of land, whether it be problems raised by the co-localization of buildings, congestion in housing settlement or environmental nuisances caused by the building of certain structures. Thus, the presence of support goods jointly consumed by different actors or their neighbours conditions the existence of a constraint of proximity.

In this perspective, geographical proximity plays a determining role in the development of tensions and conflicts between neighbours (sometimes temporary) or even between users, with their different expectations from one same area. Indeed, geographical proximity, generally presented as being *desirable*, can also be *unwelcome*. The principle is as follows:

- *Geographical proximity* is searched for by economic and social actors who need temporary or permanent proximity. They are, for example, seasonal migrants, tourists, hikers, etc, who wish to satisfy – always for short periods – a need for proximity to the countryside in the context of leisure activities. In the field of production, firms seek to locate their silos or factories of transformation close to areas of agricultural production in order to limit transport expenses and loss, or to settle near their competitors in order to benefit from their knowledge.
- However, *geographical proximity* is also often *unwelcome* by actors who are forced to tolerate a constraint of proximity (TIR and DIEHL, 2002). It is the case when secular or new land owners have to tolerate the presence of their neighbours and endure the toxic emissions and discharges provoked by the latter, or when an activity that is a source of environmental nuisance (noise, visual, olfactory pollution) is launched in their neighbourhood. This situation can also be found when users do not agree on the use of an area, which some would like to use for leisure while others wish to use it for agricultural purposes, for example, or even when access of different categories of users raises problems, when there are access restrictions, for example, or when the multi-use of the area proves difficult or even impossible.

In all these situations, *the* ambivalence between the desirable geographical proximity and the unwelcome geographical proximity generates tensions and conflicts. First, the presence of support goods conditions the existence of a constraint of proximity from which one can only escape through mobility, i.e. moving to a different area in the case of a private resident, or relocating in the case of an economic activity. But this possibility is conditioned by two factors: the availability of resources (land, underground resources, water, etc. can be found here and not anywhere else) and financial constraints, which can be an obstacle to mobility. Note that the same person can successively be in a situation of desirable proximity and then in a situation of unwelcome proximity, e.g. a hiker involved in a land-use conflict with hunters. Second, the economic and social

actors using the space are often confronted with a problem that largely conditions their relations with other actors and the solutions retained to solve the difficulties caused by the forced co-localizations: this phenomenon is called here *inequality in the face of space*. It emerges in cases of neighbourhood or contiguity. The physical characteristics of support-goods such as land or water, their very nature and the variety of situations and localizations that they impose result in the actors finding themselves in different and often unequal positions. Unlike in a relation of a productive nature, or an exchange on the local market, in this situation they cannot hope for a balanced relationship since the very topology of the space and their respective situations lead them to asymmetrical relations. The difference is at the basis of the relations between actors and cannot disappear in the context of their interactions (TORRE and CARON, 2002).

It is the case, for example, of a catchment basin in which micro localization proves essential for the treatment of sewerage or for the management of eroding runoff, the actors upstream benefit from a much more comfortable situation than those downstream who collect the discharge or must manage important volumes of water in the case of flooding, for example.

From these oppositions between unwelcome and desirable geographical proximity, tensions and conflicts emerge between neighbours, the different users of the space. These tensions and conflicts are characterized by their micro-local nature (i.e. between contiguous or close neighbours) and by the fact that they emerge in relation to the uses of space. Whether punctual or repetitive, they emerge at an inter-individual level (bad relations between neighbours, assaults, recourse to third parties, retortion, retaliation) but they can also be managed by individuals (e.g. elected people) and moral people or groups, in particular by associations representing the users of the space, administrations, local or territorial collectivities. The conflicts and tensions are not systematically solved; they can last, with phases of confrontations and periods of antagonism.

Thus, geographical proximity is a source of conflicts and presents negative dimensions seldom emphasized in the apologetic literature that often discusses it. The question that should be raised now is that of the resolution of the conflicts and of the modalities of coordination between local actors. They can be found in the other type of proximity: *organized proximity*. In this case, if the complex play between geographical proximity and organized proximity is visible, the crucial part played by the different forms of organized proximity once again proves essential.

It is important to stress first of all that organized proximity is not only a modality of resolution of conflicts and tension, but also it can generate conflicts. For example, refusing to accept the presence of certain social or ethnic classes in the more wealthy housing settlements could be likened to a symptom of the 'not

in my backyard' (NIMBY) syndrome: geographical proximity then fails because of much stronger organized proximities (CHAMBOREDON and LEMAIRE, 1970; BOSCHMA, 2003).

However, organized proximity is generally mobilized as a modality of anticipation, mediation or resolution of conflicts that emerge at the local level. Indeed, going to court – often presented in the literature as a sanction of the failure of the cooperative solution, and as being the most recognized form of conflict resolution – is but one possible solution to conflicts, or may even represent one phase of its history. Different modalities of conflict management, which sometimes can be carried out in parallel, must be distinguished (WALL and CALLISTER, 1995): negotiation, verbal or physical confrontation, the recourse to the justice system, third-party mediation or the use of the media in order to bring the matter to public opinion. It is during the negotiation phases, outside the peaks of conflictuality, and during mediations through third parties, that organized proximity intervenes.

Organized proximity contributes to the production of more or less temporary compromise, at the local level, between the actors that generate tensions or conflicts. It makes it possible either to anticipate the conflict (see COMMONS's, 1950, definition of the terms of negotiation), or to launch a mediation that will lead to a compromise or to attempt to stop the conflict by finding a permanent solution. The negotiation then depends on the rules imposed at a local level by the regional, national or supra-national authorities, but it also aims to produce rules for local use, negotiated and produced collectively by the local actors in order to manage conflict situations. It can be carried out on indirect bases, through technical acts, signs (e.g. road signs) or third parties who attempt to facilitate mediation. But it mostly takes a direct and explicit form expressed through face-to-face relations or within collective organization and consultation mechanisms.

In this case, both logics of belonging and similarity of organized proximity are required to ensure that the process of coordination functions. Belonging to one same network or one same organization enables the actors to start discussing the rules that must be produced within the implemented negotiation mechanism and the technical modalities of the solutions to be found (logic of belonging). But it is mainly the logic of similarity that is mobilized. First, because it conditions the acceptance of the common rules of negotiation that are at the origin of the process and are indispensable to its launching. Second, because it alone allows the production of collective rules accepted by all parties involved in the negotiation, and the production of beliefs and anticipations shared by all actors, a temporary and revisable compromise that surpasses conflicts and tensions and enables the actors to trace a common path. It is this construction that organized proximity

enables, e.g. in the case of the resolution of conflicts regarding the creation of a refuse incinerator or dump. Although it is not always easy to find a solution that satisfies all parties involved, sharing one's viewpoint with others and consulting with them often makes it possible to reach a satisfactory compromise.

Thus, in the case of land-use conflicts, the founding distinction between geographical proximity and organized proximity finds a new expression in which geographical proximity is a source of conflict and tension that can be solved by mobilizing the resources of organized proximity.

CONCLUSION

The objective of this paper was to pave the way for an analysis of the relations between proximity and localization of activities and people. Both notions are indeed often mistaken for one another in the contemporary literature, geographical proximity being *de facto* classified as co-localization.

The method consisted of exploiting the semantic wealth of the notion of proximity. The authors have distinguished, on the analytical plan, both types of proximity (geographical and organized) and proposed a grid of analysis of the main models of geographic organization of activities by articulating both types of proximity. Four scenarios were obtained.

The phenomenon of tension between geographical proximity and organized proximity was then introduced in order to discuss problems that are often underestimated in spatial economy. This has enabled the definition of three main lines of research for spatial economics.

First, organized proximity offers powerful mechanisms of long-distance coordination that constitute the foundation of the increasing geographic development of socio-economic interactions. The confusion made in the literature between interactions of information and knowledge exchange and the constraint of being located in proximity neglects the fact that the collective rules and representations of organizations do manage, and at a distance, an increasing part of these interactions. By defining behavioural rules and means of sharing information and knowledge, organized proximity increases the possibilities of long-distance coordination. As a matter of fact, it has been observed that globalization is accompanied by an increasing production of coordination standards, including between organizations (e.g. ISO 9000 standards). The rapid diffusion of information and communication technologies, added to this increasing formalization of coordination, contributes to increasing the potential of long-distance coordination. What matters is not so much to determine whether or not coordination requires geographical proximity to analyse how the supports of coordination, i.e. rules and technologies,

make it possible to extend so considerably the possibilities of long-distance coordination. Similarly, rather than claiming that the phenomenon of agglomeration no longer has a reason for being, it is more relevant to analyse how the foundations of the agglomeration process are evolving.

It was then shown that there was a disjunction between the need for geographical proximity and co-localization of actors, by introducing professional mobility and what has been called here temporary geographical proximity. Also emphasized was the ability of big organizations to manage the presence in different areas of their units, so as to adapt to the temporal nature of the need for geographical proximity of these units (permanent, temporary, secondary), whereas smaller organizations are more constrained by fixed co-localizations, which are only needed for certain phases of their interactions. The analysis of individuals' mobility and of its impact on the economic organization of geographical space must be developed and must not be restricted to discussing traditional aspects such as commuting or international migrations. The increase in mobility multiplies the possibilities of long-distance coordination as it provides a way of maintaining face-to-face interactions even between actors in geographically distant areas. It also reinforces the role of the nodes of transport networks as a factor of agglomeration.

Finally, the often neglected question was raised of the negative effects of geographical proximity normally praised for its relational virtues (e.g. it generates confidence, reduces costs of transaction). But in the case of limited support-goods, geographical proximity creates tensions between the actors who use them and tends to damage the local relational fabric. In this case, one could say that geographical proximity plays against organized proximity (to use the language used here). However, the negative externalities of geographical proximity can be limited by integrating them within organizations or institutions, i.e. through a re-composed organized proximity enabling one to solve conflicts and launch processes of cooperation or negotiation within ad-hoc mechanisms. A more dialectic approach to geographical proximity (positive and negative effects) and to its relation to organized proximity is thus proposed for this purpose.

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NOTES

1. The paper distinguishes mono-polar spaces, defined by the relations between one centre and one periphery, and multipolar spaces, defined by the existence of a network of poles.
2. Here, 'organization' is a term that designates any structured unit of relations. It might take any form of structure, e.g. a firm, an administration, a social network, a community and a milieu.
3. This obviously does not mean that all the beliefs of members are identical but that there is a common core of beliefs through which the organization identifies itself as one collective entity. Furthermore, the common corpus of beliefs can be based on the representation of the organization as a place of conflicts.
4. Direct externalities of geographical proximity are direct interdependencies between actors requiring geographical proximity to occur, whatever the form of externality, financial or technological.
5. The 'Proximity Dynamics' group, made of French-speaking economists, sociologists and geographers, since the early 1990s has started a collective reflection aimed at displaying convergences and coherence in the ensemble of new theoretical approaches of the economic space. This reflection lies in the common belief that space is not neutral and must be taken seriously by analysis. As soon as the group was put together, the ambition of the proximity researchers was (and still is) to explain the nature of the effects of proximity and to contribute to the endogenization of the space variable in the economic theory. Works and articles have been published by the members of this group. Among the most important are: BELLET *et al.* (1993, 1998), RALLET and TORRE (1995), GILLY and TORRE (2000), DUPUY and BURMEISTER (2003), and PECQUEUR and ZIMMERMANN (2004). In English: KIRAT and LUNG (1999), RALLET and TORRE (2000), and TORRE and GILLY (1999). Furthermore, the 'Proximity Dynamics' group organizes a Congress on Proximity every other year. The most recent congress took place in June 2004 in Marseilles, France (<http://www.139.124.177.94/proxim/program.php>).

REFERENCES

- ASHEIM B. (1996) Industrial districts as 'learning regions': a condition for prosperity?, *European Planning Studies* **4**, 379–400.
- BATHELT H., MALMBERG A. and MASKELL P. (2004) Clusters and knowledge: local buzz, global pipelines and the process of knowledge creation, *Progress in Human Geography* **28**, 31–56..
- BELLET M., COLLETIS G. and LUNG Y. (Eds) (1993) Economie de proximités, *Revue d'Economie Régionale et Urbaine* **no. 3**. [Special Issue].
- BELLET M., KIRAT T. and LARGERON C. (Eds) (1998) *Approches multiformes de la proximité*. Hermès, Paris.
- BOSCHMA R. (2003) *Proximity and Innovation. Some Critical Remarks*. Utrecht University, WP.
- CHAMBOREDON J. C. and LEMAIRE M. (1970) Proximité spatiale et distance sociale. Les grands ensembles et leur peuplement, *Revue Française de Sociologie* **11**, 3–33.
- COMMONS J. R. (1950) *The Economics of Collective Action*. University of Wisconsin Press, Madison.

- DUPUY C. and BURMEISTER A. (Eds) (2003) *Entreprises et territoires. Les nouveaux enjeux de la proximité*. La Documentation Française, Paris.
- FELDMAN M. (1999) The new economics of innovation, spillovers and agglomeration: a review of empirical studies, *Economics Innovation and New Technology* **8**, 5–25.
- FELDMAN M. and MASSARD N. (2002) *Institutions and Systems in the Geography of Innovation*. Kluwer, Dordrecht.
- FREEL M. S. (2003) Sectoral patterns of small firm innovation, networking and proximity, *Research Policy* **32**, 751–770.
- GALLAUD D. and TORRE A. (2005) Geographical proximity and the diffusion of knowledge (the case of SME's in biotechnology), in FUCHS G., SHAPIRA P. and KOCH A. (Eds) *Rethinking Regional Innovation*. Kluwer, Dordrecht.
- GERTLER M. S. (2003) Tacit knowledge and the economic geography of context, or the undefinable tacitness of being (there), *Journal of Economic Geography* **3**, 75–99.
- GILLY J. P. and TORRE A. (Eds) (2000) *Dynamiques de Proximité*. L'Harmattan, Paris.
- GRANOVETTER M. (1973) The strength of weak ties, *American Journal of Sociology* **78**, 1360–1380.
- GROSSETTI M. and BÈS M. P. (2001) Interacting individuals and organizations: a case study on co-operations between firms and research laboratories, in KIRMAN A. and ZIMMERMANN J.-B. (Eds) *Economics with Heterogeneous Interacting Agents*, pp. 123–147. Springer, Berlin.
- HOWELLS J. R. L. (2002) Tacit knowledge, innovation and economic geography, *Urban Studies* **39**, 871–884.
- HURIOT J. M. (Ed.) (1998) *La ville ou la proximité organisée*. Anthropos, Paris.
- JAFFE A. B., TRAJTENBERG M. and HENDERSON R. (1993) Geographic localization of knowledge spillovers as evidenced by patent citations, *Quarterly Journal of Economics* **108**, 577–598.
- KIRAT T. and LUNG Y. (1999) Innovation and proximity. Territories as loci of collective learning processes, *European Urban and Regional Studies* **6**, 27–38.
- KIRMAN A. (1999) Quelques réflexions à propos du point de vue des économistes sur le rôle de la structure organisationnelle dans l'économie, *Revue d'Economie Industrielle* **88**, 91–110.
- KRUGMAN P. (1991) Increasing returns and economic geography, *Journal of Political Economy* **99**, 483–499.
- LUCAS R. E. (1988) On the mechanics of economic development, *Journal of Monetary Economics* **22**, 3–42.
- MAILLAT D. (1995) Territorial dynamics, innovative milieus and regional policy, *Entrepreneurship and Regional Development* **4**, 1–20.
- MALMBERG A. and MASKELL P. (2002) The elusive concept of localization economies: towards a knowledge-based theory of spatial clustering, *Environment and Planning A* **34**, 429–449.
- ORFEUIL J. P., MASSOT H. and BELLANGER F. (2000) *Éléments pour une prospective de la mobilité*. INRETS, Paris.
- OTA M. and FUJITA M. (1993) Communications technologies and spatial organisation of multi-unit firms in metropolitan areas, *Regional Science and Urban Economics* **23**, 695–729.
- PECQUEUR B. and ZIMMERMANN J. B. (Eds) (2004) *Economie de Proximités*. Hermès, Paris.
- RALLET A. (2002) L'Economie de proximités: propos d'étape, *Études et Recherches* **33**, 11–26.
- RALLET A. and TORRE A. (Eds) (1995) *Economie industrielle et Economie spatiale*. Economica, Paris.
- RALLET A. and TORRE A. (2000) Is geographical proximity necessary in the innovation networks in the era of global economy?, *GeoJournal* **49**, 373–380.
- SCHELLING T. (1960) *The Strategy of Conflict*. Harvard University Press, Cambridge, MA.
- TETHER B. (2002) Who co-operates for innovation and why: an empirical analysis, *Research Policy* **31**, 947–967.
- TIR J. and DIEHL P. (2002) Geographic dimensions of enduring rivalries, *Political Geography* **21**, 263–286.
- TORRE A. and CARON A. (2002) Conflits d'usage et de voisinage dans les espaces ruraux, *Sciences de la Société* **57**, 95–113.
- TORRE A. and GILLY J. P. (1999) On the analytical dimension of proximity dynamics, *Regional Studies* **34**, 169–180.
- VELTZ P. (1996) *Mondialisation, villes et territoires: l'économie d'archipels*. PUF, Paris.
- WALL J. A. and CALLISTER R. R. (1995) Conflict and its management, *Journal of Management* **21**, 515–558.
- WALLSTEN S. (2001) An empirical test of geographic knowledge spillovers using geographic information systems and firm-level data, *Regional Science and Urban Economics* **31**, 571–599.
- ZIMMERMANN J. B., COLLETIS G., GILLY J. P., PECQUEUR B. and PERRAT J. (1995) *L'Ancrage territorial des activités industrielles et technologiques*. Commissariat Général du Plan, Paris.